



Statistical Analysis

BAS331 – BAS332

Assignment 5

Instructors: *Dr. Emad Yacoub*

Dr. Ashraf Fahim

Dr. Abd El Ghany Bedair

TA:

Chapter 4

Analysis of Variance – ANOVA
One Way ANOVA

1] An experiment was carried out to compare electrical resistivity for six different low-permeability concrete bridge deck mixtures. There were 26 measurements on concrete cylinders for each mixture; these were obtained 28 days after casting. The entries in the accompanying ANOVA table are based on information in the article “In-Place Resistivity of Bridge Deck Concrete Mixtures” (ACI Materials J., 2009:114–122).

(A) Complete the following one-way ANOVA table.

(B) Perform the analysis of variance for this experiment. (Use $\alpha = 5\%$)

Source of variation	D.F	S. S	M.S	F
Between Samples (Columns)		117.4		
Error			13.929	
Total		5664.415		

2] Suppose the National Transportation Safety Board wants to examine the safety of cars Type A, cars Type B and cars Type C. It collects a sample of three for each of the treatments (cars types). Using the hypothetical data provided below, test whether mean pressure applied to the driver’s head during crash test is equal for each type of cars. Use $\alpha = 0.05$

Car type A	Car type B	Car type C
643	469	484
655	427	456
702	525	402

3] Consider the following 5 random samples:

A	B	C	D	E
21	35	45	32	45
35	12	60	53	29
32	27	33	29	31
28	41	36	42	22
14	19	31	40	36
47	23	40	23	29
25	31	43	35	42
38	20	48	42	30

Perform the analysis of variance, and test the hypothesis at 0.05 level of significance, that the samples come from populations having the same mean.

4] Three sections of the same elementary statistics course are taught by 3 teachers. The final grades were recorded as follows:

A	B	C
73	88	68
89	78	79
82	48	56
43	91	91
80	51	71
73	85	71
66	74	87
60	77	41
45	31	59
93	78	68
36	62	53
77	76	79
	96	15
	80	
	56	

Is there a significant difference in the average grades given by the 3 teachers? Use $\alpha = 0.05$

5] Four laboratories are being used to perform chemical analyses. Samples of the same material are sent to the laboratories for analysis as part of the study to determine whether or not they give, on the average, the same results. The analytical results for the 4 laboratories are as follows:

Laboratory			
A	B	C	D
58.7	62.7	55.9	60.7
61.4	64.5	56.1	60.3
60.9	63.1	57.3	60.9
59.1	59.2	55.2	61.4
58.2	60.3	58.1	62.3

Perform the analysis of variance and give conclusion concerning the laboratories. Assuming that the laboratory variances are equal.

6] An investigation was conducted to determine the source of reduction in yield of a certain chemical product. It was known that the loss in yield occurred in the mother liquor; that is, the material removed at the filtration stage. It was felt that different blends of the original material may result in different yield reductions at the mother liquor stage. The following are results of the percentage reduction for 3 batches at each of 4 preselected blends:

Blend			
1	2	3	4
25.6	25.2	20.8	31.6
24.3	28.6	26.7	29.8
27.9	24.7	22.2	34.3

Is there a significant difference in the average percentage reduction in yield for the different blends? Use a 0.05 level of significant.